

# Lumi Analysis Pipeline – User Guide

## 1. Check if Python is installed

First, open Terminal / Command Prompt:

### Windows:

Click the Windows search bar, type:

```
cmd
```

and press Enter.

A Command Prompt window will open.

Run:

```
python --version
```

### macOS:

Open Spotlight Search (⌘ + Space), type:

```
Terminal
```

and press Enter.

A Terminal window will open.

Run:

```
python3 --version
```

If a Python version appears, continue to the **step 2**.

If Python is not installed, download and install it from: <https://www.python.org/downloads/>

Choose either **Windows** or **macOS** and download the latest stable version available.

### Note - **Windows**:

during installation, make sure to check:

☒ Add Python to PATH

## 2. Save the Lumi Analysis pipeline folder

Save the Lumi pipeline folder somewhere easy to access on your computer. Desktop is highly recommended for quick, easy access.

### 3. Prepare your data

Place all raw Lumi CSV files inside one folder. Either use the folder 'Analysis' Lumi automatically creates with the raw data files (recommended) or manually place them in a folder with a name of your choosing.

```
Analysis/  
├── 1a_Raw.csv  
├── 1b_Raw.csv  
├── 2a_Raw.csv  
└── 2b_Raw.csv
```

### 4. Edit the Excel file – settings.xlsx

Open the settings Excel file inside the Lumi Analysis pipeline folder. Some settings are marked with (\*) - these settings **MUST** be edited or verified before the first run. All other settings may be adjusted later after an initial run if needed.

The settings file contains several columns, only the blue one is editable. Detailed descriptions of each setting is provided in the Excel file. Every time you change the settings file, you must click Save + re-run the pipeline from the terminal (**step 6**).

Notes – correct formats for the settings:

- Leaving a cell empty is equivalent to the value:

None

- A list must be written using square brackets: [ ]  
Names have to be inside quotes – "example". Numbers and Boolean values (True/False) should **NOT** be in quotes. Examples:

```
[1, 2, 3]  
["control", "Dex", "Dex 25"]  
[True, "sample", 2, False]
```

- The syntax of the folder paths depends on the operating system. For example, the use of backslashes (/ or \) between folders:

**Windows:**

```
C:\Users\YourName\Desktop\Analysis
```

**macOS:**

```
/Users/YourName/Desktop/Analysis
```

## 5. Open Terminal inside the Lumi folder

### Windows:

Right-click inside the Lumi folder and choose:

Open in Terminal

### macOS:

Right-click the Lumi folder and choose:

New Terminal at Folder

Or navigate manually:

```
cd /path/to/Lumi_folder
```

Example:

```
cd /Users/YourName/Desktop/Lumi
```

## 6. Run the pipeline

### Windows:

```
python -m scripts.run_pipeline
```

### macOS:

```
python3 -m scripts.run_pipeline
```

Note: the first run of the pipeline may take slightly longer, as required packages will be installed automatically. During this process, the terminal may display installation messages similar to the example below:

```
Downloading matplotlib-3.10.9-cp314-cp314-win_amd64.whl (8.3 MB)
 8.3/8.3 MB 7.8 MB/s 0:00:01
Using cached contourpy-1.3.3-cp314-cp314-win_amd64.whl (232 kB)
Using cached cyclex-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.63.0-cp314-cp314-win_amd64.whl (2.3 MB)
 2.3/2.3 MB 7.9 MB/s 0:00:00
Using cached kiwisolver-1.5.0-cp314-cp314-win_amd64.whl (75 kB)
Downloading packaging-26.2-py3-none-any.whl (100 kB)
Using cached pillow-12.2.0-cp314-cp314-win_amd64.whl (7.2 MB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, packaging, kiwisolver, fonttools, cyclex, contourpy, matplotlib
 7/8 [matplotlib]
```

## 7. Find your results folder

Results will be saved to the output folder specified in the settings file. If a full path to the folder was provided, the results would be in that location. If only a folder name was provided instead of a full path, it would most likely be created inside the Lumi pipeline folder.